

PBMC3K Cell Annotation Investigation Context

Objective

Investigate differences in Leiden clustering and marker gene identification across multiple preprocessing and differential expression runs performed on the PBMC3K dataset.

The goal is to document:

- Processing choices made in each run
- Marker gene results
- Cluster sizes
- Questions that arose during analysis
- Technical errors encountered

No final conclusions are included.

Questions Raised During Analysis

HVG Usage

The Scanpy PBMC3K tutorial:

```
sc.pp.highly_variable_genes(adata)
```

calculates HVGs but does not subset the dataset:

```
adata = adata[:, adata.var.highly_variable]
```

Instead, all genes remain in the AnnData object.

Question raised:

- Why are HVGs calculated but not used for subsetting?
- How should one decide whether to use only HVGs or retain all genes?

Observation:

The tutorial later uses:

```
sc.tl.rank_genes_groups(  
    adata,  
    "leiden",
```

```
mask_var="highly_variable",  
method="wilcoxon"  
)
```

which restricts differential expression testing to genes marked as highly variable.

Run 1

Processing

- Doublet detection performed
- Doublets removed
- Ribosomal gene filtering done
- Marker genes generated from clustered dataset

Cluster Sizes

```
Cluster 0: 296  
Cluster 1: 401  
Cluster 2: 768  
Cluster 3: 327  
Cluster 4: 469  
Cluster 5: 153  
Cluster 6: 153  
Cluster 7: 29
```

Top Marker Genes

Rank	0	1	2	3	4	5	6	7
1	CCL5	IL32	RP11-879F14.2	CD74	LYZ	GZMB	AC009501.4	RP11-511B23.1
2	NKG7	LTB	GPRC5C	HLA-DRA	S100A9	NKG7	RP11-219B4.7	NCKAP1
3	B2M	B2M	C5AR2	HLA-DRB1	S100A8	GNLY	AC195454.1	PAK3
4	GZMA	IL7R	CEACAM4	HLA-DPB1	CST3	CTSW	PLD6	USP30
5	CST7	GIMAP7	LY6G6F	HLA-DPA1	FTL	PRF1	FCGR3A	OBSCN

Additional Top Markers for Cluster 7

Rank	Gene
1	RP11-511B23.1
2	NCKAP1
3	PAK3
4	USP30
5	OBSCN
6	CTA-250D10.23
7	DRAXIN
8	RP11-164H13.1
9	KCNG1
10	PDZD4

Run 2

Processing

- Doublet detection skipped
- Same workflow otherwise
- Marker genes recalculated

Top Marker Genes

Rank	0	1	2	3	4	5	6	7
1	CCL5	CD74	LTB	LYZ	NKG7	RP11-879F14.2	AC009501.4	RP11-511B23.1
2	NKG7	HLA-DRA	IL32	S100A9	GNLY	C5AR2	RP11-219B4.7	NCKAP1
3	IL32	HLA-DRB1	B2M	S100A8	GZMB	GPRC5C	IL6	PAK3
4	B2M	HLA-DPB1	CD2	CST3	CTSW	LY6G6F	MTRF1	USP30
5	GZMA	HLA-DPA1	TMSB4X	FTL	PRF1	ITGB3	PLD6	OBSCN
6	CST7	HLA-DQA1	MALAT1	FCN1	HKDC1	AGO4	ZNF528	RP11-493L1.1
7	CTSW	CD79A	IL7R	TYROBP	RP13-131K19.7	CEACAM4	PAK3	DRAXIN

Rank	0	1	2	3	4	5	6	7
8	MALAT1	HLA-DQB1	GIMAP7	LGALS2	SEMA4F	FAM3D	C1orf186	RP11-164H
9	GZMK	CD79B	TRAF3IP3	FTH1	AC073072.5	QPRT	PLEK2	KCNG1
10	VPS13A	CD37	AQP3	AIF1	ZBTB32	GBGT1	BACE2	PDZD4

Run 3

Processing

Changes relative to previous runs:

- Doublet detection skipped
- Ribosomal gene filtering skipped
- Followed tutorial more closely
- HVGs calculated but dataset not subset
- Differential expression recalculated

Top Marker Genes

Rank	0	1	2	3	4	5	6	7	8
1	CCL5	CD74	LTB	MALAT1	LYZ	GNLY	LST1	HLA-DPB1	SPARC
2	NKG7	CD79A	IL32	CD27	S100A9	NKG7	FCER1G	HLA-DPA1	GNG11
3	B2M	HLA-DRA	IL7R	SELL	S100A8	GZMB	AIF1	HLA-DRA	PPBP
4	IL32	CD79B	CD2	NDFIP1	TYROBP	PRF1	COTL1	HLA-DRB1	PF4
5	GZMA	HLA-DPB1	GIMAP7	IL7R	FTL	CTSW	FCGR3A	HLA-DQA1	SDPR
6	CST7	HLA-DQA1	AQP3	LTB	FCN1	GZMA	IFITM2	CST3	CD9
7	CTSW	MS4A1	ANXA1	LDLRAP1	CST3	CST7	FTH1	FCER1A	GPX1
8	GZMK	HLA-DRB1	VIM	MAL	LGALS2	B2M	IFITM3	CD74	HIST1H2AC
9	LYAR	HLA-DQB1	B2M	GIMAP5	S100A6	FGFBP2	SAT1	HLA-DRB5	RGS18

Rank	0	1	2	3	4	5	6	7	8
10	MALAT1	CD37	GIMAP5	TRAF3IP3	FTH1	CD247	FTL	HLA-DQB1	NRGN

Open Questions

1. Why did PPBP appear among top markers in Run 3 but not in run 1 and run 2 ?
2. Did the HVG set change between runs?
3. How does skipping doublet removal alter cluster composition and marker ranking?
4. How does skipping ribosomal gene filtering influence HVG selection and downstream marker discovery?